

XINYUAN (ELLA) LI

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SUMMARY

Data Science and Business Economics student at UC San Diego (GPA 3.99/4.0) with production machine learning and full-stack AI experience: built real-estate valuation models at a proptech startup and shipped an LLM tutoring extension now used by 400+ students. Targeting 2027 summer internships in data science and analytics.

EDUCATION

University of California, San Diego | *Halicioğlu Data Science Institute*

Expected June 2028

B.S. Data Science and Business Economics · GPA 3.99/4.0 · Provost Honors, all eligible quarters

Relevant Coursework: Probabilistic Modeling and Machine Learning, Introduction to Data Visualization, Data Structures & Algorithms for Data Science, Theoretical Foundations of Data Science I & II, Practice of Data Science, Intro to Data Management, Applied Statistical Data Analysis, Econometrics A/B/C, Microeconomics A, Linear Algebra, Calculus (Multivariable)

EXPERIENCE AND RESEARCH

Student Researcher, AI in Education | *Data Science Teaching and Learning Lab, UC San Diego*

Oct 2025 – Present

- Built and shipped a full-stack JupyterLab extension (React/TypeScript frontend, Python/Tornado backend) that embeds an AI tutor directly in student notebooks, deployed to around 400 students across 2 quarters of DSC 10.
- Designed a context-aware retrieval layer that sanitizes live notebook state — active cell, markdown instructions, and outputs — and surfaces relevant lecture examples via TF-IDF retrieval with Gemini reranking, reducing generic/off-topic tutor responses by grounding answers in course-specific material.
- Shipped exam mode, topic-based practice retrieval, and Socratic prompting constrained to course-specific APIs (babypandas), logging around 88,000 tutoring sessions to date.

Student Researcher, Economics | *Department of Economics, UC San Diego*

Jun 2026 – Present

- Estimated how UC faculty summer teaching supply responded to the \$10,000-per-course pay cap introduced in 2019, applying regression discontinuity and difference-in-differences to 34,585 administrative course records.
- Built the full cleaning-to-estimation pipeline in Stata and ran robustness checks across bandwidths and specifications; results feed into a working paper.

Data Science Tutor | *Halicioğlu Data Science Institute, UC San Diego*

Sep 2025 – Present

- Owned an automated grade-reporting pipeline (Python, bash, Gradescope API) that computes final scores across labs, homework, quizzes, exams, and participation — handling slip days, drops, and curving — then pushes personalized reports to more than 200 students weekly, replacing an error-prone manual process and ensuring consistent, reproducible grading.
- Maintained the course website and GitHub repository, releasing weekly labs, midterm and final projects on schedule across 3 quarters.
- Led discussion sections, pod meetings, office hours, and one-on-one exam reviews for more than 200 students per quarter on regression, permutation testing, and Python in Jupyter.

Data Scientist Intern | *IDX Exchange*

Jun 2025 – Sep 2025

- Engineered more than 50 features from 125,542 real-estate transaction records (CoreLogic Trestle API), using exploratory analysis to isolate the price drivers that shaped the final model architecture.
- Trained and tuned linear regression, random forest, and XGBoost valuation models, reducing MAPE from 20.9% to 0.85% through iterative tuning (log-transforming price, pruning low-value features) by over the existing baseline at a final R^2 of 0.86; presented results weekly to senior data scientists.
- Documented the end-to-end modeling pipeline and set reproducibility standards — environment pinning, seeded splits, and model versioning — adopted by the team.

PROJECTS

Pokémon Team Recommender | *Python, Streamlit · DS3 DataHacks 2025*

Apr 2025

- Built a Streamlit application that recommends optimized six-Pokémon teams from user stat preferences through a custom weighted scoring algorithm over 1025 Kaggle records; presented to judges at DataHacks 2025.

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn, XGBoost), R, SQL, Stata, TypeScript/JavaScript, Java, HTML/CSS

Machine Learning and Statistics: Regression modeling, feature engineering, A/B testing, permutation testing, causal inference (difference-in-differences, regression discontinuity), exploratory data analysis

Tools: Git/GitHub, JupyterLab, PostgreSQL, React, Tornado, Streamlit, Matplotlib, Seaborn, Excel (advanced), Cursor

Spoken Languages: English (fluent), Mandarin (native)

LEADERSHIP, HONORS, AND AWARDS

Orientation Leader | *Warren College, UC San Diego*

Apr 2025 – Present

- Presented campus resources to more than 300 incoming students and synthesized post-event survey data from respondents to reshape the following year's programming.

Events Board Member | *Warren College Student Council, UC San Diego*

Sep 2024 – Sep 2025

- Planned and executed campus events for more than 500 attendees, managing a \$27,581 budget across 20 events.

Provost Honors, Warren College, UC San Diego (UC GPA ≥ 3.5 , 12+ graded units)

Fall 2024 – Spring 2026

Fourth Place, Programming Competition, Women in Computing, UC San Diego

2025